A Gymnasium Substrate and World-Model Baseline for ARC-AGI-3:
DreamerV3 Fits the World but Cannot Yet Act in It

Hasaan Ahmad | Supervisor: Soumya Banerjee (University of Cambridge)

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1 Motivation and gap

ARC-AGI-3 is a benchmark of 25 public interactive abstract-reasoning games in which the rules and goals are *not* given and must be inferred through active play. It is scored by RHAE (Relative Human Action Efficiency): per level, $s_i = \min\big((\text{human\_actions}/\text{ai\_actions})^2,\, 1.15\big)$; a game score is the level-index-weighted mean $\sum_i s_i\, i / \sum_i i$ over all levels (including uncompleted ones), and the total score averages games equally. The ARC-AGI living survey [2] reviews ∼80 papers across ARC-AGI-1/2/3, documents a consistent ∼3× performance cliff from ARC-AGI-1 to ARC-AGI-3, finds only 3 of 80 papers report any ARC-AGI-3 result, *zero* of them world-model based, and explicitly names world-model induction as the required next step. Ying et al. [3] argue the same from first principles: today's evaluations test static pre-trained representations, whereas the human capability of interest is using an abstract world model as a prior to build a task-specific one on the fly, and they propose ARC-AGI-3-style games as the ideal testbed. Lee et al. [1] give the one direct precedent — DreamerV3 beating PPO on multi-step ARC-AGI-1 tasks — and explicitly call for scaling to harder interactive tasks. This work is the first model-based RL (MBRL) entry on ARC-AGI-3 and supplies the missing standard RL entry point.

2 Contributions

(1) Substrate (primary artifact). The first Gymnasium- and DreamerV3-`embodied`-compatible wrapper for ARC-AGI-3. It exposes a 4102-way flat discrete action space (0–4→`ACTION1--5`; 5–4100→`ACTION6` over the 64×64 click grid via `unravel_index`; 4101→`ACTION7`/undo) with per-game boolean action masking applied to the actor logits at reset; a $(64, 64, 3)$ `uint8` image observation in DreamerV3's native modality; an offline loader for the 340-replay human-demonstration corpus; and a post-hoc RHAE harness. Round-trip action coding, full-corpus loader parse, and RHAE are property-/regression-tested (318-test suite green). DreamerV3 is used *stock and unforked* (config `size12m`, the ARC-AGI-1-precedented choice [1]); the wrapper registers as an `embodied` environment with no upstream modification.

(2) Controlled negative result. A paired baseline of stock DreamerV3 on 6 public games × 2 seeds, evaluated from-scratch (cold) vs. warm-started from a cross-game world model pretrained on all 340 human replays mixed across 25 games (Regime B: shared WM, per-game actor). Cross-game WM pretraining yields no measurable benefit within seed variance.

(3) Mechanistic diagnosis. The world model's reconstruction and dynamics losses collapse to convergence while RHAE stays ≈0: the model *learns to predict the environment*, but the controller cannot exploit it inside imagination within a 500k-step budget.

3 Method

World model (encoder + RSSM + decoders + reward/continue heads) was pretrained self-supervised on the full mixed 340-replay buffer (Phase 3, 1×A100; image reconstruction loss 53.66→0.14, dynamics loss monotone then flat at ∼1.12). For each game the pretrained WM is loaded as initialisation, a fresh actor and critic are created, the per-game human replays are pre-populated, and online interaction proceeds for 500k env-steps with default DreamerV3 hyperparameters. The from-scratch (cold) arm is identical except the warm-start flag is removed — so warm-start is the *only* difference between arms, giving a clean ablation of the central Regime-B premise. Human baselines for RHAE use the upper-median of first-time-player action counts; levels with < 2 completers are dropped from both numerator and denominator (resulting human-baseline coverage 70.49%, 129/183 levels over 25 games).

4 Results

Only the easiest game, vc33, is ever non-zero. On vc33 the per-seed sign of Δ disagrees (s0: warm +0.0137; s1: cold +0.0016); with $n = 2$ the warm−cold delta lies inside per-seed variance, so the Regime-B hypothesis that cross-game WM pretraining contributes to downstream control is *not supported*. The pre-registered Phase-4 gate (RHAE > 0 on $\geq 2/3$ of {vc33, sb26, cd82}) failed at 1/3. This is a clean, paired, 6-game negative result, not an unlucky single run.

Table 2 is the central finding in one view. cd82's world model fits *tighter* than vc33's (image loss 0.06 vs. 0.16) — yet vc33 is the only game the controller ever solves, while cd82's policy never leaves a 1.000 random-action

Table 1: Paired RHAE @ 500,000 env-steps. Stock DreamerV3 `size12m`, 6 public games × 2 seeds, cold (from-scratch) vs. warm (cross-game WM-pretrained). Δ = warm−cold. Higher is better; human-parity ≈ the per-level 1.15 cap.

	Cold		Warm			
Game	s0	s1	s0	s1	Cold $\bar{x}$	Warm $\bar{x}$
vc33	0.0411	0.0182	0.0548	0.0166	0.0297	0.0357
sb26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
cd82	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
tn36	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ls20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
lf52	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Combined (mean of 6)					**0.0049**	**0.0059**

Table 2: The dissociation: the world model fits *better* on a game the controller never solves (cd82) than on the only game it ever solves (vc33). Both columns are seed-0 warm runs from the same paired experiment as Table 1; the pattern holds across all 6 pilot warm cells.

Signal	vc33 (only solved)	cd82 (never solved)
WM image-recon. loss	58.3 → 0.16 (−99.7%)	31.0 → **0.06** (−99.8%)
WM dynamics loss	10.6 → 1.24	11.4 → 1.03
Train eps. clearing ≥1 lvl	17/9751	0/4950
Eval eps. clearing ≥1 lvl	4/23	0/24
Random-action frac. @500k	0.750	**1.000**
Post-hoc RHAE	0.0548	0.0000

fraction across the entire 500k-step run. The model learns the environment; the controller cannot exploit it. This holds across all six pilot warm cells ({vc33, sb26, cd82}×2 seeds): image loss collapses >95% everywhere, while both gate-failure games clear 0/24 eval episodes in every cell. The bottleneck is the controller, not the world model — not a single lucky or unlucky run.

5 Discussion and positioning

The result is the first MBRL data point on ARC-AGI-3 and is consistent with, and sharpens, the survey's ∼3× cliff [2]: a model that provably fits these environments still cannot act in them under a stock controller and a realistic compute budget, which is exactly the "static-representation vs. on-the-fly task model" gap Ying et al. predict [3]. It also qualifies the Lee et al. [1] precedent: DreamerV3's ARC-AGI-1 success does *not* transfer to the interactive, rule-inference regime of ARC-AGI-3 at this scale, and cross-game WM pretraining does not close the gap — isolating the controller, not the world model, as the locus of difficulty. **Limitations:** a compute-bounded 500k-step budget and $N = 6$ games × 2 seeds — a scoped claim about *stock DreamerV3 at this budget ± this pretraining*, not that MBRL cannot do ARC-AGI-3; the diagnosis, not the headline number, carries the weight. **Status and planned work.** The substrate (contribution 1) is complete, tested, and reusable independent of any result; the negative result and its diagnosis (2–3) are established across the full 6-game paired sweep. The immediate next step is a quantitative confirmation of the controller-vs.-model localisation — linear probes on the RSSM latents and generative-rollout fidelity (FVD) — followed by a targeted test of whether the gap closes under controller-side changes alone (training ratio and exploration schedule) while keeping the world model stock. The substrate contribution does not depend on the outcome of either.

References

[1] D. Lee et al. *Reinforcement Learning and Model-Based Reasoning on the Abstraction and Reasoning Corpus* (DreamerV3 vs. PPO on ARC-AGI-1). arXiv:2408.14855, 2024.

[2] *The Abstraction and Reasoning Corpus: A Living Survey* (review of ∼80 papers across ARC-AGI-1/2/3; documents the ∼3× cliff; zero world-model approaches on ARC-AGI-3). arXiv:2603.13372, 2026.

[3] L. Ying et al. *Instance vs. Abstract World Models: Evaluating Dynamic World-Model Induction* (proposes ARC-AGI-3 as the testbed). arXiv:2507.12821, 2025.

[4] D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap. *Mastering Diverse Control Tasks through World Models* (DreamerV3). Nature, 2025. arXiv:2301.04104.

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